

Exp-1:

Read an Image and Convert to Grayscale

Code:

```
import cv2, matplotlib.pyplot as plt
img = cv2.imread('image.png')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
plt.subplot(121); plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)); plt.axis('off')
plt.subplot(122); plt.imshow(gray, cmap='gray'); plt.axis('off')
plt.show()
```

Output:



Exp-2:

Gaussian Blur

Code:

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread('image.png')
blur = cv2.GaussianBlur(img, (5,5), 0)
plt.subplot(121); plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)); plt.axis('off')
plt.subplot(122); plt.imshow(cv2.cvtColor(blur, cv2.COLOR_BGR2RGB)); plt.axis('off')
plt.show()
```

Output:



Exp-3:

Canny Edge Detection

Code:

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread('image.png')
edges = cv2.Canny(cv2.cvtColor(img, cv2.COLOR_BGR2GRAY),100,200)
plt.subplot(121); plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)); plt.axis('off')
plt.subplot(122); plt.imshow(edges, cmap='gray'); plt.axis('off')
plt.show()
```

Output:



Exp-4:

Histogram Equalization

Code:

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread('image.png')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
eq = cv2.equalizeHist(gray)
plt.subplot(121); plt.imshow(gray, cmap='gray'); plt.axis('off')
plt.subplot(122); plt.imshow(eq, cmap='gray'); plt.axis('off')
plt.show()
```

Output:



Exp-5:

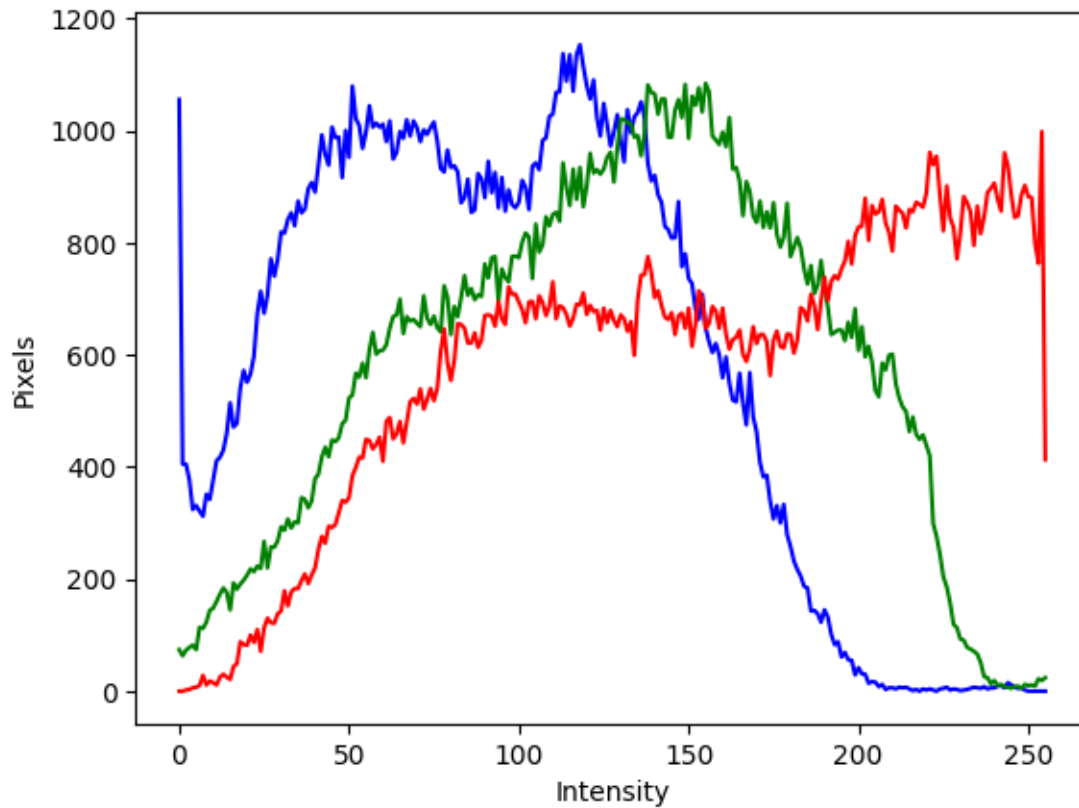
Histogram Analysis

Code:

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread('image.png')
colors = ('b','g','r')
for i, col in enumerate(colors):
    hist = cv2.calcHist([img],[i],None,[256],[0,256])
    plt.plot(hist, color=col)
plt.xlabel('Intensity')
plt.ylabel('Pixels')
```

plt.show()

Output:



Exp-6:

Image Erosion using Erode Function

Code:

```
import cv2
import matplotlib.pyplot as plt
import numpy as np
img = cv2.imread('image.png')
kernel = np.ones((5,5), np.uint8)
eroded = cv2.erode(img, kernel, iterations=1)
plt.subplot(121); plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB)); plt.axis('off')
plt.subplot(122); plt.imshow(cv2.cvtColor(eroded, cv2.COLOR_BGR2RGB)); plt.axis('off')
plt.show()
```

Output:

